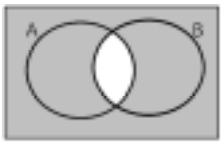
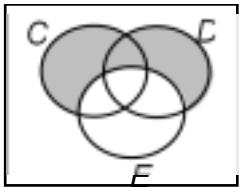
Question	Answer	Marks	Partial Marks
16(a)	Multiples of 3 Prime numbers + | 3 | **6** | 9 2 | 5 | **8** | 11 3 | 6 | **9** | **12** **5** | **8** | **11** | **14**	**2**	**B1** for at least 4 correct entries
16(b)	$\frac{2}{5}$ oe	**2**	**B2FT** for $\frac{their\,2}{their\,5}$ or **B1FT** for $\frac{their\,2}{k}$ k is any integer in the range $1 \leqslant k \leqslant 7$ or $\frac{c}{their\,5}$ c is 0, 1 or 2
17	$\frac{3}{5}$ oe and $-\frac{7}{2}$ oe	**1**	
18	$x^2 - 11x + 24$ [= 0] or $y^2 - 16y + 39$ [= 0]	**M2**	**M1** for $x^2 - 9x + 21 = 2x - 3$ oe or $y = \left(\frac{y+3}{2}\right)^2 - 9\left(\frac{y+3}{2}\right) + 21$ oe
	$(x - 8)(x - 3)$ [= 0] or $(y - 13)(y - 3)$ [= 0]	**M1**	or for correct factors for *their* quadratic equation or for correct use of quadratic formula for *their* equation
	[x =] 3 [y =] 3 [x =] 8 [y =] 13	**B2**	**B1** for one correct pair or two correct x values or two correct y values. If B0 scored **and** at least 2 method marks scored **SC1** for correct substitution of both of *their* x values or *their* y values into $y = x^2 - 9x + 21$ or $y = 2x - 3$
19	 	**2**	**B1** for each
20(a)	32	**2**	**M1** for f(6) = 8 or $\text{ff}(x) = 2^{\left(2^{x-3}\right)-3}$ oe

Question	Answer	Marks	Partial Marks
20(b)	$x + 21$	**1**	
20(c)	–1	**2**	**M1** for $\frac{1}{16}$ oe or 2^{-4} oe
21	$2x^3 - 7x^2 - 12x + 45$ final answer	**3**	**B2** for unsimplified expansion of the three brackets with at most one error or for simplified four-term expression of correct form with three terms correct or **B1** for correct expansion of two of the given brackets with at least three terms out of four correct
22	196.6 or 196.60… and 343.4 or 343.39…	**3**	**B2** for one correct angle or **M1** for $\sin x = -\frac{2}{7}$ or better If 0 scored **SC1** for two angles that sum to 540°
23	$\frac{3y-5}{2(x-12)}$ or $\frac{3y-5}{2x-24}$ final answer	**4**	**SC3** for answer $\frac{3y-5}{x-12}$ or **B3** for $(3y - 5)(x + 12)$ **and** $2(x - 12)(x + 12)$ or $(2x - 24)(x + 12)$ or **B2** for $(3y - 5)(x + 12)$ or $2(x - 12)(x + 12)$ or $(2x - 24)(x + 12)$ or $(2x + 24)(x - 12)$ or **B1** for $3y(x + 12) - 5(x + 12)$ or $x(3y - 5) + 12(3y - 5)$ or $2(x^2 - 144)$ or $(x - 12)(x + 12)$